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Structured illumination for 3-D Mie imaging and 2-D attenuation measurements in optically dense sprays

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Abstract

Qualitative description of spray structure and geometry in the near-field region of atomizing sprays is particularly challenging due to the lack of visibility through such optically dense media. Similarly, quantitative determination of the optical depth can be particularly underestimated as a result of detecting photons which have experienced a number of scattering events. In this paper we demonstrate, based on average imaging, the possibility to reconstruct the "true" 3-D Mie representation of an aerated spray, where image artifacts introduced by multiple scattering have been suppressed by means of *Structured Laser Illumination Planar Imaging* (SLIPI). Here, six images were used, instead of three usually employed in structured illumination, to reconstruct a single SLIPI image. Based on a similar approach, a 2-D estimation of optical depth has been performed and compared with the conventional single-point attenuation measurement.
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1. Introduction

Planar laser imaging, also known as laser sheet imaging, is a commonly used technique for two-dimensional characterization of combusting two-phase flows [1]. Due to its whole-field detection, small depth of field and simplicity, various approaches based on planar imaging have been developed during the past three decades for both qualitative and quantitative purposes. In the field of spray diagnostics, the technique has been widely employed to describe spray structure/geometry, measure the cone angle and estimate spray penetration. In parallel, quantitative determinations of flow vectors with Particle Tracking Velocimetry [2], liquid volume fraction with Planar Laser Induced Fluorescence [3] and droplet size by means of interferometric planar imaging [4] and Planar Droplet Sizing [5,6], have been areas of strong activities.

However, despite these numerous advantages, laser sheet imaging of sprays suffers from three main limitations when applied within the intermediate single-to-multiple scattering regime, where the single scattering approximation is no longer valid [7,8]. This is problematic since the

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intermediate scattering regime covers most optical diagnostics of industrial and fuel spray systems (i.e. $1 < OD < 10$). The first limitation corresponds to the exponential attenuation of the laser sheet intensity as it propagates through a cloud of droplets, due to scattering and absorption along the incident direction. This well known phenomenon, called *laser extinction*, is described by the Beer–Lambert law. Secondly, the induced signal, regardless of the scattering process involved – elastic or inelastic – is reduced while traveling towards the camera, also according to the Beer–Lambert relation. This process is referred to as *signal attenuation* or *signal trapping*. Finally, depending on the optical thickness of the spray, photons can experience several scattering events within the sample prior to detection, a process known as *multiple scattering*.

Different approaches either to reduce or to compensate for the two first sources of error have been investigated in the past. Simultaneous or sequential bidirectional excitation schemes have been utilized to reduce some of the losses caused by laser extinction [9–11]. Furthermore, by taking advantage of the *a priori* decay of intensity, iterative computer algorithms have also been employed to compensate for laser extinction [12,13]. Accounting for signal attenuation is more complicated as it may require three-dimensional measurements, as discussed in [14]. Sick and Stojkovic [10] calculated the local error by estimating the signal attenuation between the laser sheet and the camera.

In spite of efforts to improve the accuracy of the local signal and even though multiple scattering is considered as the major source of error [6,8], no satisfying experimental solution has been offered to account for multiple scattering issues (on a planar configuration) since the first utilization of *Structured Laser Illumination Planar Imaging* (SLIPI) [15,16].

In this article the possibility of extracting a succession of planar Mie images, based on the SLIPI technique, at different depths within the dense region of an aerated water spray is demonstrated. By using such a scanning approach, a 3-D Mie reconstruction of the spray has been performed and results between the conventional and SLIPI technique are compared. In the second part of the paper, a novel approach to extract the optical depth in 2-D using structured illumination is demonstrated and the accuracy is compared with the traditional attenuation point measurement. It is believed that the combination of 3-D imaging and 2-D extinction measurements is the key approach to ultimately allow all three main sources of errors to be corrected for.

2. Structured laser illumination planar imaging

SLIPI is a novel imaging technique which has demonstrated great potential for multiple scattering suppression in planar imaging of scattering media, strongly improving visualization [15–18]. The fundamental principle of the technique (based on *structured illumination* [19,20]) is to superimpose a sinusoidal intensity modulation, in the spatial domain, on a homogeneous laser sheet. This modulation pattern acts as a "finger print" that allows the singly scattered photons to be recognized, as they will scatter accordingly. In contrast, photons which deviate from their original trajectory will lose the modulation pattern and appear as a position-dependent intensity noise on the recorded image. These photons are the multiply scattered ones. To only collect the single scattering intensity, the method consists in extracting the amplitude of the modulated component in the recorded image. However, by using a single intensity modulated laser sheet, parts of the probed sample are not illuminated (see Fig. 1). This is compensated for by shifting the line-pattern vertically, inducing a shift on the position of singly scattered photons (while the contribution of multiply scattered light remains mostly unaffected). When vertically shifting the modulation $(n-1)$ times, n images are recorded with an angular shift of $\Phi = 2\pi/n$ between each image. Finally, the resulting *SLIPI image* is constructed from n modulated images (where $n \geqslant 3$) according to:

$$SLIPI = \frac{\sqrt{2}}{n}\left(\sum_{j=1}^{n-1}\sum_{k=j+1}^{n}(I_j - I_k)^2\right)^{1/2}, \qquad (1)$$

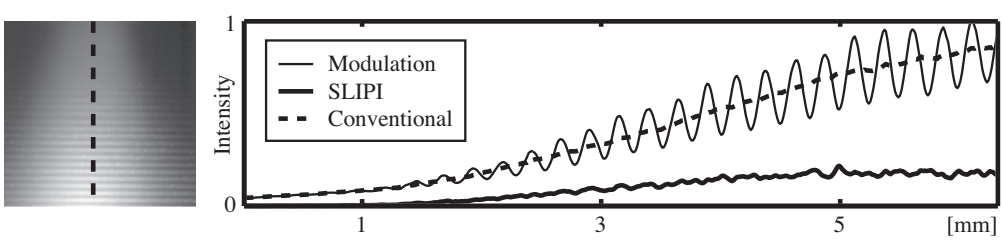

Fig. 1. Left side: Example of a single planar image using a modulated laser sheet. Right side: Cross-sections for such a single modulated, together with the conventional and SLIPI results. Notice the non-uniform suppression of diffuse light (difference between conventional and SLIPI).

where I_j and I_k are intensity values and the subscript j and k denotes the different images. In this equation, the pair-wise subtraction $(I_j - I_k)$ removes similar features introduced by the diffuse light while unique features from the direct light are kept. By instead averaging the n images, as described in Eq. (2), the *conventional image* is reconstructed.

$$Conv. = \frac{1}{n}\sum_{j=1}^{n} I_j \tag{2}$$

This image is, in theory, equal to that acquired when using a homogeneous (non-modulated) laser sheet, thus allowing the two techniques to be easily compared, as illustrated in Fig. 1. The construction of the SLIPI image requires a minimum of $n = 3$ vertically shifted images, which is the approach used in previous studies [15–18]. Increasing the number of images leads to a shorter line movement. Although time consuming, this improves the effectiveness of the suppression, especially for scattering media containing strong vertical variations. Another issue that may be improved when using a shorter line movement, concerns the presence of residual line structures. These artifacts arise when the contribution of directly scattered light differs between the intensity modulated images. Spatial changes in the laser profile may also give rise to such unwanted effects. Methods to reduce these residuals are described in [21] but may not always be sufficient. By using a larger number n of images, the presence of these artifacts can be statistically reduced. For these reasons, six modulated images, instead of three, have been used in this article to reconstruct both the SLIPI and conventional images.

3. Experimental set-ups

3.1. SLIPI-Mie for 3-D image reconstruction

The laser source consisted of a frequency doubled Nd:YAG laser ($\lambda = 532$ nm) operating at 10 Hz. As structured illumination is sensitive to beam inhomogeneities, a small portion of the central gaussian beam was selected and spatially filtered to create a near top-hat intensity distribution. The line structures were formed using a transmission Ronchi grating (10 lp/mm) and projected into the spray as a thin (slightly diverging) sheet of light using a combination of cylindrical lenses (see Fig. 2). To avoid unwanted interference between the Ronchi frequencies, all but the two 1st order frequencies were physically blocked. For each SLIPI image a sequence of six vertically shifted images, each averaged over 1650 accumulations, were recorded. A 14-bit electron multiplying CCD camera (iXon-897, 512×512 pixels) was used to record the light scattered at 90°. To keep the contribution of multiple scattering as low as possible, a small detection acceptance angle ($\theta_a \sim 1°$ as suggested in [22]) was used by choosing a f-number of $f/\#22$, on a Nikon 105 mm focal length objective. The imaged area was situated right beneath the nozzle tip and corresponds to a squared area of the dimension 6.2×6.2 mm^2. To obtain a 3-D Mie representation of the spray, 11 different planes were imaged by translating the nozzle 250 μm between each recording. As a result, half of the spray could be reconstructed in 3-D, within a total volume of $6.2 \times 6.2 \times 2.5$ mm^3. For the *OD* measurements, the camera position was changed and imaged a screen upon which the transmitted light was guided.

The nozzle used during the experiment was a Delavan AL-45 of tip diameter Ø = 3 mm, producing a narrow angle spray pattern with a nominal angle of 15–20° in ambient air. This air assisted nozzle, is an internal mixing nozzle, where a change in liquid or air injection pressure results in a change of either flow or atomization, respectively. Here, the investigated spray was running in steady state at 3 bar water pressure and 2.8 bar air pressure delivering a flow rate of 0.85 l/min in ambient air.

3.2. Attenuation measurements in 2-D

The optical depth *OD*, also called optical distance [23] or optical thickness [24] is defined as:

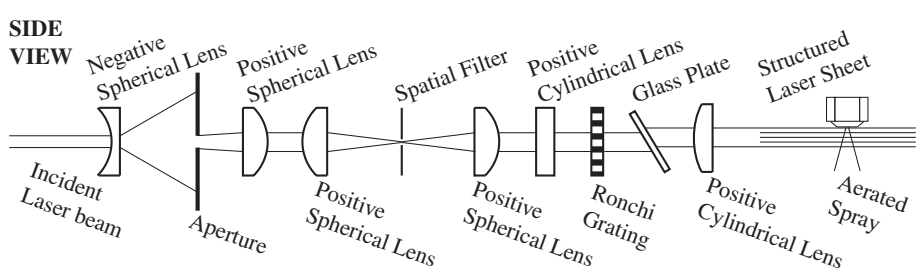

Fig. 2. Description of the optical arrangement. A flat intensity profile is extracted by spatially filtering the central part of the incident laser beam. To create a modulated intensity a Ronchi grating is used. By tilting a glass plate the required vertical shift of the line pattern is obtained. For the 3-D measurements the camera was situated at 90° and the illumination was focused into a thin sheet. For the *OD* measurements the camera imaged a screen on which the transmitted light was guided.

$$OD = \sigma_e \cdot N \cdot l \quad (3)$$

where σ_e is the extinction cross-section, N is the number density of particles and l is the length crossed by a beam of light along its incident direction. OD is the exponential power of the Beer–Lambert law and corresponds to the average number of scattering and/or absorption events occurring along the incident direction when photons propagate through a turbid medium. Experimentally, the optical depth is measured from the light attenuation I_f/I_i according to:

$$OD = -\ln\frac{I_f}{I_i} \quad (4)$$

where I_i and I_f are the incident and transmitted light intensity, respectively. The major source of error in the experimental measurement of the optical depth is the intensity contribution arising from multiply scattered light, leading to an underestimation. The easiest way to reduce the multiple scattering intensity is to use a small detection acceptance angle. Common practice is to measure the intensity ratio I_f/I_i using a single laser beam. For more advanced techniques combinations of time-gating, spatial and polarization filtering are used to inhibit the detection of scattered light [25,26].

In addition to the 3-D measurements using SLIPI, this article also presents a new technique to measure optical depth, where the ultimate aim is to obtain such information two-dimensionally with both high spatial resolution and accuracy. To evaluate the effectiveness of this new method, which is based on structured illumination, the optical depth was also estimated using two other methods. Figure 3 shows a summary of the different measurement schemes.

In the first method (1) the spray was illuminated with a top-hat beam profile, optically expanded to cover the entire measurement volume. Due to the large field of illumination and the imaging nature of the system (inducing a larger detection acceptance angle), this method is strongly affected by multiple scattering. Therefore, this measurement should be considered merely as a demonstration of how the detection of such undesired light degrades the accuracy. By inserting a Ronchi grating into the beam path, a sinusoidal intensity modulation is created, which is the basis for the new and proposed method (2). By vertically shifting this induced line structure and applying Eq. (1), multiply scattered light is diminished. To fully evaluate the effectiveness of (2) two different illumination approaches, based on this method, were tested (see Fig. 3); in (2.1) the entire measurement volume was covered, while in (2.2) the beam width was reduced into a 1-D sheet.

The last method (3) was a traditional attenuation point measurement where a small laser beam was guided through the spray. The detector, a photodiode, was situated at 7.4 m from the spray, resulting in a detection acceptance angle of 0.025°. The measurement was performed at 6 mm below the orifice, where a minimum signal from the center of the spray could still be recorded by the photodiode. Note that for methods (1), (2.1) and (2.2), the iXon camera was used to image the incident and transmitted light, which were guided onto a screen. The main objective with this investigation was to determine whether approaches (2.1) and/or (2.2) could provide sufficient accuracy, compared to method (3). For all the presented measurements the response of the camera or the photodiode has been assumed to be within a linear regime.

4. Results

4.1. SLIPI-Mie for 3-D image reconstruction

To reconstruct the 3-D surface of the aerated spray a shape-based interpolation scheme [27,28] was implemented on the SLIPI and conventional images. The routine is based on binary data and creates additional image planes, filling the gaps between the original ones. Before applying a threshold (to convert the data into binary information) all images were first normalized followed by histogram adjustment to assure that at least 1% of the data was saturated at intensities 0 and 1. A threshold of 0.2 was then used on all images.

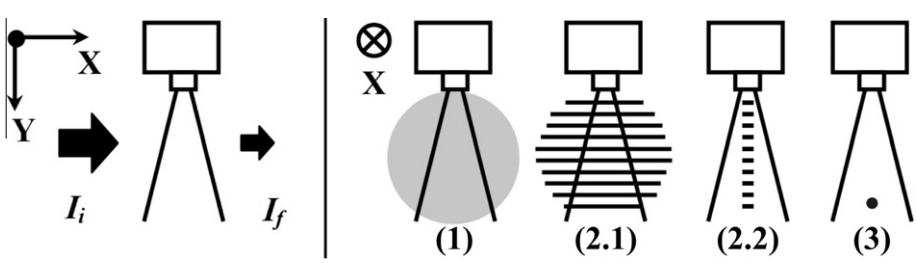

Fig. 3. Illustration of the various light transmission measurements. (1): Flat beam illumination, (2.1): structured flat beam illumination, (2.2): modulated 1-D sheet illumination and (3): single "point source" illumination.

However, when probing different depths within the spray, image intensities will differ due to signal attenuation and laser extinction. For this reason, such a single *static* threshold should in principle be avoided. To extract a more accurate *dynamic* value these sources of error should ultimately be treated prior to 3-D reconstruction. Note that the present post-processing will strongly reduce the visual influence of signal attenuation.

Conventional and SLIPI results of the 3-D Mie representations are presented in Fig. 4 (orifice situated at $(x, y, z) = (0, 0, 0)$). Here, 6 of the 11 images used to generate the 3-D representation are shown in the top of the figure. Despite the high optical depth of the spray, the conventional images lacks the appearance of laser extinction, in contrast to the SLIPI images. Based on this observation, it is believed that the signal in the conventional images is dominated by multiply scattered light, while the detected intensity in the SLIPI images arise mostly from single scattering. When probing at $Z = 0$ mm (along the central axis), the SLIPI intensity tends to zero after crossing only half of the spray. This is not induced by an absence of droplets, but because almost no ballistic (non-scattered) light penetrate this far. As a result, an irregular "void structure" appears in the central region of the spray when reconstructed in 3-D (shown in Fig. 4). For conventional imaging, the 3-D reconstruction exhibits a plain and more symmetrical structure. As a conclusion, the 3-D SLIPI-Mie provides a full three-dimensional view of how far into the spray the incident light penetrates before its intensity is reduced below the threshold value. These results further demonstrates both the need for compensation methods as well as the need to suppress multiply scattered light prior to applying such methods.

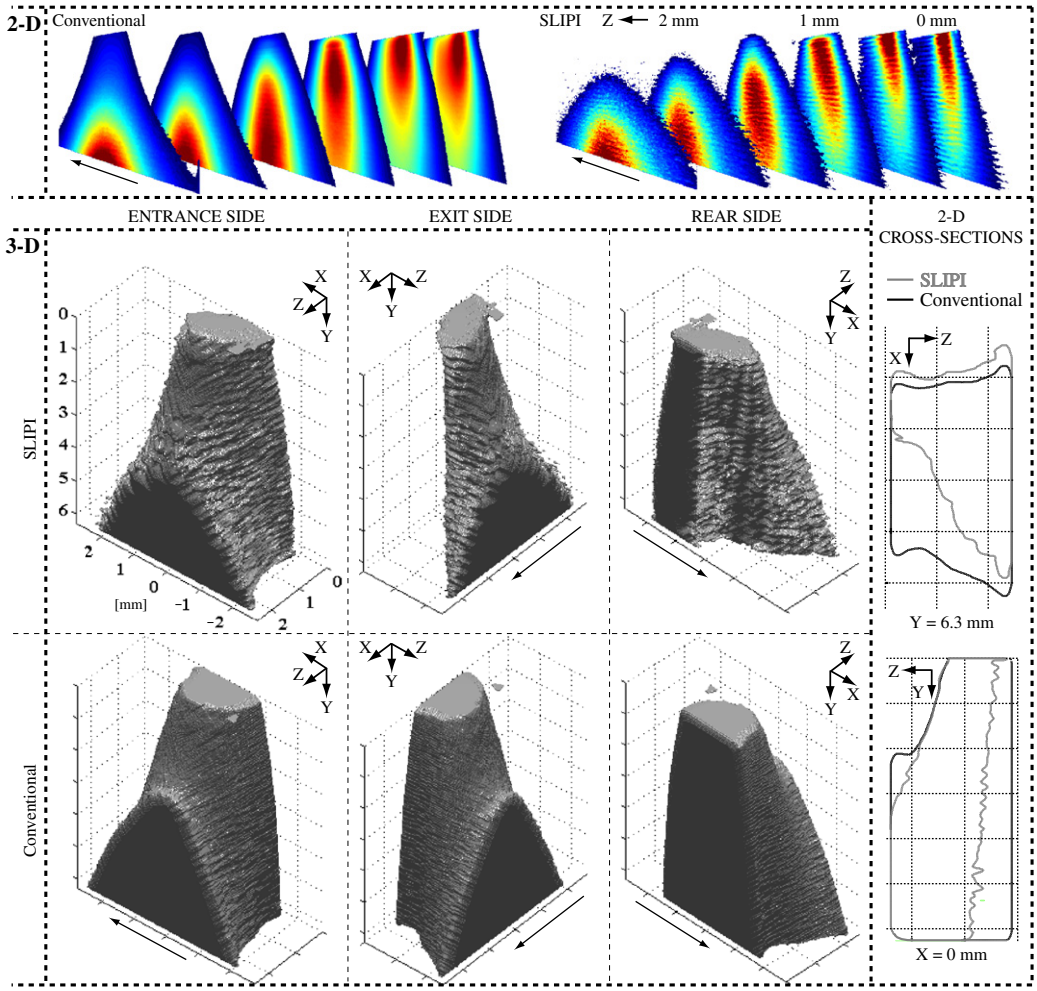

Fig. 4. 2-D and 3-D imaging of the aerated spray for both SLIPI and conventional imaging, together with 2-D cross-sections. The arrows indicate the direction of the incident light. Spray orifice situated at $(x, y, z) = (0, 0, 0)$.

4.2. Attenuation measurements in 2-D

The results from the methods described in Section 3.2, are presented in Fig. 5. As expected, by using method (1) the optical depth is underestimated, with values ranging between $0 < OD < 5$. This is in contradiction to the strong reduction of light with distance observed in the 3-D measurements. By modulating the flat beam profile and applying Eq. (1), method (2.1) provides a much wider range, $0 < OD < 8$, of optical depth. With method (2.2) the estimated value of optical depth is even further increased. Differences between (2.1) and (2.2) are highest close to the orifice; however, OD values start to coincide at 5–6 mm downstream (see magnified image in Fig. 5). These results indicate that method (2.1) is able to provide results equivalent to method (2.2) at $OD \leqslant 7$. This upper limit of (2.1) is due to the finite dynamic range of the camera as both the spray boundaries (where the droplet number density is low) as well as the densest region were probed simultaneously. When instead applying (2.2), where only the central region was illuminated, the incident laser intensity could be increased without saturating the camera. From these observations it is believed that discrepancies between (2.1) and (2.2) mostly come from dynamic range issues rather than multiple scattering effects. Still, to increase the accuracy of both these approaches, the suppression of multiple scattering must be further improved. One possibility is to increase the modulation frequency, as demonstrated by Berrocal et al. in [18].

Finally, measurement (3) shows the highest estimation of OD with a maximum value of $OD = 8$. Contrary to the imaging techniques where the detection acceptance angle θ_a was set to 1°, θ_a was reduced to 0.025° in method (3), providing a better rejection of diffuse light. Note, however, that due to both the forward scattering nature of the scattering phase function of the water droplets and the high optical thickness of the spray, the experimental values of OD remain still underestimated even with method (3). This issue is experimentally shown and discussed in [25,29], and numerically demonstrated in [7].

Measuring optical depth is of fundamental interest in many different fields of research, e.g. absorption measurements for determining species concentrations. Knowledge of OD can also be used to compensate for losses of signal due to laser extinction, as demonstrated by Hertz and Aldén [12]. However, using the traditional attenuation point measurement provides merely a single value of OD and to obtain either 1-D or 2-D information would require a scanning process. The main objective with the current investigation is therefore to develop and evaluate new techniques to eventually measure optical depth in 2-D. Combining such methods with 3-D imaging, as presented in Section 4.1, could potentially allow existing correction approaches to be implemented, ultimately compensating for the three sources of error associated with laser sheet imaging of scattering and absorbing media.

5. Conclusion

Different implementations of structured illumination has been tested and evaluated for the optical characterization of an aerated spray. The first set of experiments aimed at rebuilding a 3-D structure of the spray by means of several averaged SLIPI-Mie images, taken at an interval dis-

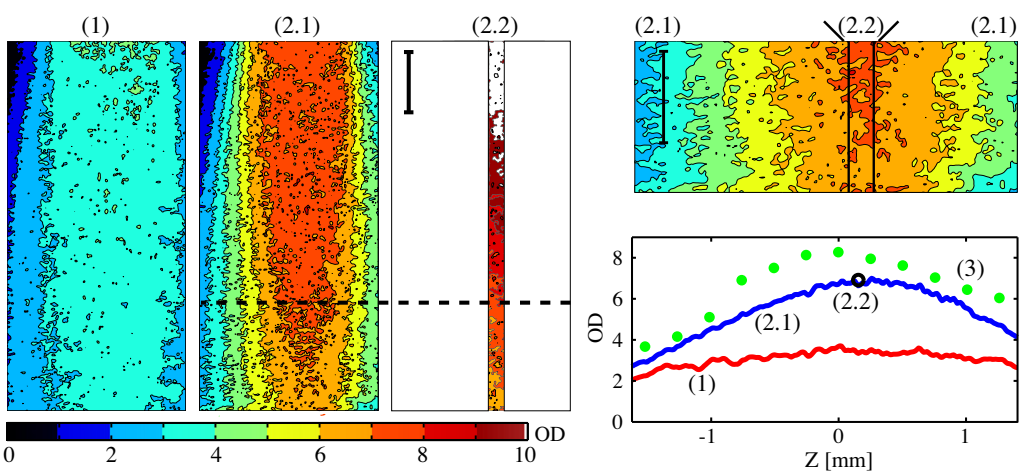

Fig. 5. Estimation of the optical depth in 2-D: Each image is associated with the illumination methods described in Fig. 3. Methods (1), (2.1) and (2.2) are based on imaging techniques while the point measurement technique (3) was performed at 6 mm from the nozzle tip, at various locations along the Z-axis. It is seen in the magnification given in the top-right corner that methods (2.1) and (2.2) tends to equal values when $OD \leqslant 7$.

tance of 250 μm. These images were created using six intensity modulated laser sheets instead of only three, usually used in structured illumination. It was found that this increase in number of images, reduced the appearance of residual lines in the final SLIPI images.

The second set of experiments consisted in evaluating a new method to obtain information of optical depth with high spatial resolution in 1-D and 2-D. It is demonstrated that this technique provides results comparable to those acquired using the single point measurement technique. Suggestions to improve the techniques involve the use of a higher modulation frequency as well as polarization filtering. By combining these results – from 3-D Mie imaging and 2-D extinction measurements based on structured illumination – correction procedures for laser extinction and signal attenuation effects can ultimately be performed. Such an approach would offer the possibility of extracting the extinction coefficient in 3-D, largely improving quantitative measurements of droplets number density and size within dense atomizing sprays.

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